

XAI: DALL-EVAL: Probing the Reasoning Skills and Social Biases of Text-to-Image Generative Models

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eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

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<https://interpretable-ml-class.github.io/>

DALL-EVAL: Probing the Reasoning Skills and Social Biases of Text-to-Image Generative Models

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(Cho, Zala, and Bansal 2023)

🍷 The Text-to-Image Landscape ---

DALL-E



DALL-E^{Small} and minDALL-E



🍷 The Text-to-Image Landscape ---

Stable Diffusion



X-LXMERT



🍷 Text-to-Image Evaluations ---

Image Quality

Whether the generated images look similar to images from training data.

- **Metrics:** Inception Score (IS) and Fréchet Inception Distance (FID)

Image Quality

Whether the generated images align with the semantics of the text descriptions.

- **Metrics:** R-precision, BLEU, CIDEr, Semantic Object Accuracy (SOA)

Image Quality ---

Metrics for evaluating Image Quality use the features of a pretrained image classifier such as Inception v3 to measure the diversity and visual reality of the generated images.

- **Inception Score:**

$$\exp(\mathbb{E}_{\mathbf{x}} \text{KL}(p(y|\mathbf{x}) || p(y)))$$

- **Fréchet Inception Distance:**

$$d^2((\mathbf{m}, C), (\mathbf{m}_w, C_w)) = \|\mathbf{m} - \mathbf{m}_w\|_2^2 + \text{Tr}(C + C_w - 2(CC_w)^{1/2})$$

🍷 Image-Text Alignment ---

Current metrics for assessing Image-Text Alignment are based on retrieval, captioning, and object detection models.

- **R-precision:** precision at the R-th position in the ranking of results for a query that has R relevant documents
- **BLEU and CIDEr:** caption-based similarity scores
- **Semantic Object Accuracy (SOA):** object detection-based alignment



🍷 Measuring Bias ---



🍷 Problem Statement ---

There is a lack of **comprehensive evaluation metrics** for text-to-image generative models like DALL-E.

🍷 Contributions: 2 Areas to Evaluate ---

1. **PaintSkills:** A compositional diagnostic dataset and evaluation toolkit.
2. **Social bias evaluation** for text-to-image generation models.



PaintSkills Overview ---

- Goal: evaluate visual reasoning of text-to-image models
 - ▶ Need 1: define “skills” that reflect visual reasoning
 - ▶ Need 2: select dataset to evaluate the visual reasoning
- PaintSkills addresses both needs
 - ▶ *dataset* and *evaluation toolkit* that evaluates visual reasoning skills for text-to-image models

Skills ---

- ➊ **Object Recognition** — given a text describing a specific object class (e.g., an airplane), a model generates an image that contains the intended class of object
- ➋ **Object Counting** — given a text describing M objects of a specific class (e.g., 3 dogs), a model generates an image that contains M objects of that class
- ➌ **Spatial Relation Understanding** — given a text describing two objects having a specific spatial relation (e.g., one is right to another), a model generates an image including two objects with the relation

VQA/GQA Shortcomings ---

- VQA/GQA: $\langle \text{image, question, answer} \rangle$ tuples
- Dataset bias
 - ▶ Skewed distribution towards few common objects, questions, and answers
- PaintSkills controls for bias between input text and objects

Approach ---

Generates text-image pairs by:

- ① Define scene configs
 - Ⓐ ensure objects, counts, and relations are uniformly distributed
- ② Generate text prompts from scene config
 - Ⓐ mention object, count, and spatial relations
- ③ Generate image from scene config
 - Ⓐ Unity simulator

🍷 Scene Config to $\langle \text{text}, \text{image} \rangle$ ---

Scene Config

- 15 MS COCO Classes: {person, dog, ...}
- Object count range: {1, 2, 3, 4}
- Spatial relations: {above, below, left, right}
- 13 backgrounds

Text: templated string

Image: Unity 3D simulator



🍷 Dataset Examples ---



Dataset Metrics ---

	Train	Test
Object Recognition	23,250	2,325
Object Counting	21,600	2,160
Spatial Relation Understanding	13,500	2,700

Evaluation Overview ---

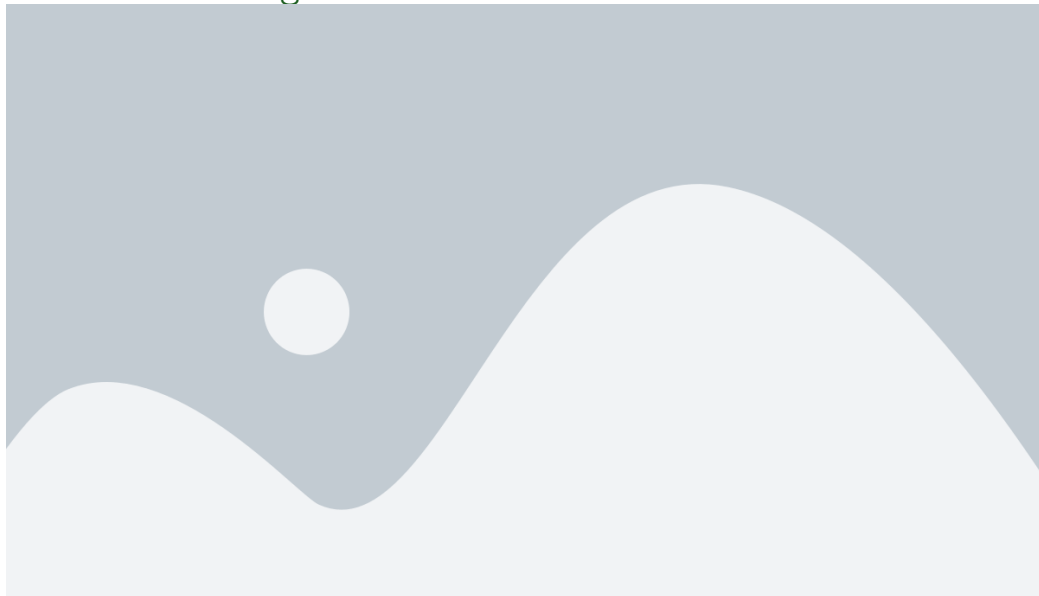
Evaluation is done on two **new** criteria:

- ① visual reasoning skills
- ② social biases

... and two **current** criteria:

- ③ image-text alignment
- ④ image quality

🍷 Visual Reasoning Skill Evaluation ---



Visual Reasoning Skill Evaluation ---

Skills are evaluated based on how well an object detector (**DETR**) can detect the object described in the input text

- Trained on MS COCO 2017 train split

Skill Accuracy Metrics

- **Object Recognition:** average accuracy whether DETR correctly identifies the target class from N generated images
- **Object Counting:** average accuracy whether DETR correctly identifies M objects of the target class
- **Spatial Relation Understanding:** average accuracy whether DETR correctly identifies both target object classes and pairwise spatial relations (above/below/left/right from 2D coordinates)

🍷 Social Bias Evaluation ---



- **Gender** refers to **sex** (male/female), not gender identity — due to human error and potential amplification of stereotypes by assigning gender labels
- Skin tone uses **Monk Skin Tone** scale: continuous color mapped to 10 skin tones

Social Bias Evaluation ---

- Bias is based on skew of distribution from images generated from **neutral** prompts
- Using 4 categories of words: profession, political, object, and other
 - ▶ e.g., “a photo of a [X]” where X is from profession/political/object/other words
 - ▶ e.g., “a person with a [X]”
- For each prompt, generate 9 images with stochastic sampling, then estimate gender/skin tone

Social Bias Evaluation ---

Automated Gender Detection

- **CLIP:** choose most prominent gender category from images with 2 classifier prompts: "a photo of a male", "a photo of a female"

Automated Skin Tone Detection

- Detect skin pixels based on RGBA and YCrCb colorspace, take average of skin pixels and match with MST skin tone

Human Evaluation

- 5 MTurkers to select gender; ask an expert for skin tone

Social Bias Evaluation ---

Measuring Bias

Obtain distributions for gender/skin tone; bias w.r.t. degree of the skewed distribution is measured using:

- **standard deviation**
- **mean absolute deviation**

of normalized counts of the gender or skin tone category

Results

Evaluated Models ---

X-LXMERT

Cross-modal transformer and a GAN-based image decoder

DALL-E style (DALL-E^{Small} and minDALL-E)

Discrete VAE that encodes images and a multimodal transformer that learns the joint distribution of text and image tokens

Stable Diffusion (v1.4)

Latent diffusion model with cross-attention conditioning

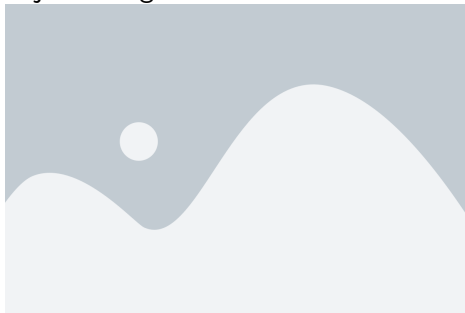
🍷 Skill Accuracy Results ---



🍷 Zero-shot Results ---

All models do not achieve high accuracy ($< 50\%$)

Only exception: **Stable Diffusion's** object recognition skill



🍲 Results after Fine-tuning ---

1. Fine-tuning improves accuracy of all models on all three skills
2. minDALL-E significantly outperforms Stable Diffusion on spatial skill

Simply scaling data and model size does not always provide better visual reasoning skills



🍷 Social Bias Results ---



🍷 Social Bias Results ---

Stable Diffusion has a stronger tendency to generate images of a specific gender or skin tone from neutral prompts than minDALL-E





Limitations ---

- Pretrained models for evaluations do not guarantee robust evaluation of text-to-image generation models trained on unseen data
- More biases could be explored beyond gender and skin tone
- More complex reasoning skills (3D spatial relations) needed
- Focuses on English-heavy datasets; more work needed on other languages

Conclusions ---

- 1 Authors propose **two new evaluation aspects** of text-to-image generation: visual reasoning skills and social biases
- 2 Introduce **PAINTSKILLS**: a dataset and evaluation toolkit measuring object recognition, object counting, and spatial relation understanding
- 3 Recent text-to-image models perform better at **object recognition** than counting and spatial relations — wide gap to upper bound remains
- 4 Models learn **gender/skin tone biases** from web image-text pairs

Discussion Questions ---

- ① Is focusing on procedurally generated data (like PaintSkills) the right path for evaluating text-to-image generative models?
- ② How do you make sure your classifiers aren't biased to begin with?
 - ▶ Evaluation of biases is dependent on unbiasedness of evaluators
- ③ How might we go about evaluating social biases beyond sex and skin tone?
- ④ Do you believe that PaintSkills can accurately assess the visual reasoning capabilities of generative models?

Thank You!

References --- |

Cho, Jaemin, Abhay Zala, and Mohit Bansal. 2023. "DALL-EVAL: Probing the Reasoning Skills and Social Biases of Text-to-Image Generative Models." In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*.